The COR Strategy: A New Method To Improve The Dip Trip Traders' Profitability

Jose Tamele[1], Wei Hao Khoong[2]
jtamele5@gmail.com[1], khoongweihao@hotmail.com[2]

Ritabrata Bhattacharyya[3]
ritabrata.bhattacharyya@wqu.edu[3]

Abstract

Traditional dip-buying strategies rely heavily on indicators like Relative Strength Index (RSI), Bollinger Bands, and moving averages to time entries, but they fail to measure the actual velocity and strength of the subsequent bullish pullbacks. To bridge this gap, we introduce in this study a novel, physics-inspired technical indicator: the Coefficient of Restitution (COR) indicator. Adapting the physics principles of collision dynamics – where the coefficient of restitution measures kinetic energy retention after impact – the COR indicator is mathematically defined as the ratio of the absolute differences between asset and market RSI values before and after price dips, offering a quantitative gauge of pullback strength relative to broader market conditions. Using a rigorous backtesting framework with walk-forward optimization across multiple timeframes (from 30-minutes to daily intervals), we evaluated the COR strategy on Magnificent Seven stocks and leveraged ETFs, incorporating volatility-adjusted position sizing via Average True Range (ATR). Our findings show that the COR strategy significantly outperforms traditional mean-reversion frameworks, particularly on high-volatility assets. Notably, at the 1-hour timeframe, the COR indicator combined with the ATR sizing achieved 41.56% total return, an 85.71% win rate, and a Sharpe Ratio of 1.51. These results provide market practitioners with a robust quantitative framework to validate rebound strength and assets trade quality prior to execution.

Keywords*: Swing trading; dip trip strategy; Coefficient of Restitution; COR indicator; COR Strategy;*

1. Introduction

Swing trading constitutes a vital segment of quantitative asset management, capturing short-to-medium-term market inefficiencies by holding positions across horizons of several days to weeks. Within this paradigm, mean-reversion and structural dip-buying strategies have gained popularity, systematically exploiting temporary price depreciations at established support levels within broader macroeconomic uptrends. However, practitioners tend to struggle to timing appropriate entry spots and resulting in severe drawdowns or capital stagnation during periods of extended market corrections. Achieving enhanced risk-adjusted returns requires a mathematically sound entry timing approach derived through robust, multi-dimensional validation metrics.

Conventionally, systematic swing traders rely on a deterministic composite of classic technical indicators – principally the Relative Strength Index (RSI), Bollinger Bands, and Moving Averages – to map structural entry points. While this traditional baseline strategy

can identify oversold conditions, it possesses a fundamental architectural limitation: it fails to quantify strength and velocity of the subsequent bullish pullback. This blind spot arises because conventional metrics evaluate asset momentum in isolation based entirely on historical, idiosyncratic price series. Consequently, they ignore broader market-wide dynamics and systematic risk factors. Such a structural omission prompts a critical quantitative question: *When an entry signal is triggered by conventional mean-reversion parameters, how can practitioners reliably quantify the underlying momentum of the bullish pullback to ensure it possesses sufficient kinetic energy to breach immediate resistance zones?*

To handle this limitation, this paper introduces the Coefficient of Restitution (COR) strategy, based on a novel, physics-inspired technical indicator. The conceptual framework of the COR indicator is mathematically derived from Newtonian collision mechanics – specifically the classic bouncing ball problem [1] – where the coefficient of restitution measures the conservation of kinetic energy and velocity retention post-impact. In physical systems, this coefficient dictates the maximum rebound height achievable after a collision. Translating this principle into financial econometrics, the COR indicator operates as an advanced filtering layer alongside traditional signal generators. While conventional indicators establish the preliminary trade architecture, the COR indicator dynamically assesses position quality by validating the systemic strength of the price rebound relative to broader market momentum.

Consequently, the COR framework provides traders with an ex-ante risk management tool to verify trade viability before capital allocation. The remainder of this study is organized as follows: Section 2 delineates the underlying theoretical framework; Section 3 details the empirical methodology and walk-forward optimization framework; Section 4 presents the backtesting results and performance metrics across the Magnificent Seven and leveraged ETFs; and Section 5 discusses the implications of the findings for algorithmic execution.

2. Theoretical Framework

Our work targets Project Track 7 (Medium/Long-Term Trading Models), Topic 1 (Swing Trading), to apply the COR indicator to the dip trip strategy, and seek to use it as a metric to determine the quality of the trading actions for the strategy. The proposed pipeline specifies Bollinger Bands, moving averages, RSI for entry timing, COR indicator for position quality estimation, volatility-adjusted stops and targets for exits.

We thus aimed to convert the physics inspired metric alongside traditional financial metrics for universe selection, parameter optimization and trading decisions across US equities and potentially other liquid asset classes, like leveraged ETFs, to maintain target profitability thresholds.

2.1. Technical Analysis

Technical analysis has a long empirical record with recent studies noting instances of economic profitability across speculative markets, but results are mixed and highly sensitive to testing choices, data-snooping control, and cost estimations, thus motivating rigorous out-of-sample validation and robust inference procedures in design and evaluation [2]. Foundational computational work also formalized indicator-based rules and inference challenges, reinforcing the need for careful model selection, validation, and statistical testing when deploying moving average, Relative Strength Index (RSI), and band-based swing strategies.

2.1.1. Relative Strength Index (RSI)

Created by Donald J. Wilder, the Relative Strength Index (RSI) is defined as a "*[...] momentum oscillator that estimates price changes by measuring how quickly and in which direction prices move*" [3]. The RSI values' band ranges from 0 to 100, where values below 30 form the oversold zone and the others above 70 constitute the overbought region.

The calculation of RSI requires the use of a predefined time window, say, a 14-day period. Along such a window:

1) Calculate the average of positive returns (APR)

$$AVR = \frac{Number\ of\ positive\ returns\ in\ the\ window}{14} \quad (1)$$

2) Calculate the average of negative returns

$$ANR = \frac{Number\ of\ negative\ returns\ in\ the\ window}{14} \quad (2)$$

3) Calculate the smoothed average gain (AG):

$$AG = \frac{(AVR \times 13 + Current\ Positive\ Return)}{14} \quad (3)$$

4) Calculate the smoothed average loss (AL):

$$AL = \frac{(ANR \times 13 + Current\ Negative\ Return)}{14} \quad (4)$$

5) Calculate the Relative Strength (RS):

$$RS = \frac{AG}{AL} \quad (5)$$

6) Calculate the RSI:

$$RSI = 100 - \frac{100}{1+RS} \quad (6)$$

In buying dip strategy, the RSI indicator is likely to trigger an entry signal if it is below 30 and the asset price is in bullish pullback.

2.1.2. Bollinger Bands

Bollinger Bands place a volatility-based formulation around a moving average by adding and subtracting a multiple of the rolling standard deviation, and the moving averages are simple arithmetic means over a fixed lookback window N applied to the asset price. For the Bollinger Bands, the midline is an N-period simple moving average, and the upper/lower bands are that average plus/minus K standard deviations, where typical choices are $N = 20$ and $K = 2$ [4] in many charting conventions.

2.1.3. Moving Averages

A simple moving average (SMA) over a window N is the arithmetic mean over the last N prices. At time t, this is given by

$$SMA_t = \frac{1}{N} \sum_{i=0}^{N-1} P_{t-i} \quad (7)$$

where P_t is the price of the asset at time t.

For the SMA, a smaller window N makes SMA more responsive to short-term price changes, while a larger N makes SMA less responsive to changes in the series of asset prices.
In swing trading the buying dip entry signal is likely to be generated when a larger window SMA crosses its smaller window version, however such a signal is subject to confirmation by other indicators.

2.2 Volatility-aware Risk Management: Average True Range

Inspired by the belief "*[...] that the range is directly proportional to volatility and it is indicative of trend*" [5], J. Welles Wilder created the Average True Range (ATR), a non-directional tool that predicts trends using securities' prices.

The calculation of ATR is a two-step procedure as follows.

Step 1 - Calculate the true range:

$$TR = max(H - L, |H - C|, |L - C|), \quad (8)$$

where H is the highest high, L is the lowest low, and C is the previous close price.

Step 2 - Calculate the Average True Average:

$$ATR = \frac{(N-1)\times Previous\ ATR + TR}{N} \quad (9)$$

with N being the length of the lookback window which is commonly set to 14 periods.

Along the algorithm we use the ATR to generate volatility-adjusted position sizes whose mathematical expression is:

$$PS = \frac{Available\ Cash \times Risk\ per\ trade}{C \times ATR} \quad (10)$$

where C is the previous close price.

2.2.1 Volatility Measures & Performance

A central result involving scaling exposure inversely to recently realized volatility (among volatility-managed portfolios) is that it increases Sharpe ratios and generates large alphas across market and factor strategies because changes in volatility are not matched by proportional changes in expected returns [6]. In particular, related work showed that momentum and similar trend strategies suffer crash risk in high-volatility episodes, which inspired volatility-targeting or scaling as a risk-management overlay [7], which is an idea closely aligned with our project's volatility-aware sizing and exits. At the same time, out-of-sample and market-specific evidence indicates that the efficacy of volatility management can attenuate or fail in certain contexts, underscoring the importance of sensitivity estimation, regime segmentation, and external validity checks for dip-trip implementations.

Our proposed use of Average True Range (ATR) for price movement volatility assessment [8] as a price-scaled range-based volatility proxy for entry conditioning and stop and target calibration is standard in practice and consistent with our project's algorithmic blueprint, and references to dip-search strategies using volatility bands. Realized volatility computed from high-frequency or daily returns provides a complementary measure shown to relate to cross-sectional momentum dynamics at the industry level, supporting the dual-proxy approach for sensitivity analysis at trade entry.

Combining Bollinger Bands with short-term oversold filters (e.g. RSI) and trend filters (e.g. moving averages), which we refer to as *traditional strategy*, reflect common swing heuristics designed to trade on localized weakness within broader uptrends, consistent with the proposed entry logic and parameter grid. The literature cautions, however, that profitability claims for technical rules can be sample-dependent and subject to selection bias, reinforcing the need for the project's walk-forward framework and explicit performance thresholds in training before out-of-sample testing [2].

Volatility-adjusted stops, trailing profit targets, and risk-per-trade position sizing align with evidence that expected returns are not linearly compensating for volatility spikes, making scaled exposure and volatility-aware exits valuable for improving realized Sharpe and drawdown characteristics in trend-following strategies [6]. Our proposed dynamic stop distance via ATR multiples and volatility-adjusted profit targets translate this insight into concrete trade management mechanics for the dip-trip strategy.

2.3. The COR strategy for buying dips

Formally, the Coefficient of Restitution (COR) in Physics' bouncing ball problem is defined by

$$e = \frac{|v_a|}{|v_b|} = \frac{relative\ speed\ before\ collision\ (separation)}{relative\ speed\ after\ collision\ (approach)} \qquad (11)$$

where if $e = 1$ then the collision is perfectly elastic (i.e. no kinetic energy is lost) and if $e = 0$ the collision is perfectly inelastic (i.e. no rebound) [1]. The numerator $|v_a|$ would consist of metrics in the present time and the denominator $|v_b|$ would consist of metrics at a particular time in the past.

The COR buying dip strategy, as developed in this work, combines the traditional buying dip approach and the COR indicator. While the traditional strategy will simply tell the trader what the optimal entry spot or price is, the COR indicator, denoted by the symbol e, provides information on the strength of the bullish pullback movement of the asset price relative to the market movement.

Now, let e denote the COR threshold, RSI_V^S denote the RSI of the variable V (either an asset or market index) at a time step S. The threshold e is thus mathematically given by

$$e = \frac{\left|RSI_{asset}^{before} - RSI_{market}^{before}\right|}{\left|RSI_{asset}^{after} - RSI_{market}^{after}\right|} \qquad (12)$$

where RSI_{asset}^{before} is the RSI of the asset before the dip, RSI_{market}^{before} is the RSI of the market before the asset's dip, RSI_{asset}^{after} is the RSI of the asset after the dip, and RSI_{market}^{after} is the RSI of the market after the asset's dip.

Inspired by the Physics Coefficient of Restitution where it indicates the amount of kinetic energy remaining in the bouncing ball right after the collision, the COR indicator has a value band ranging from 0 to 1, meaning $0 \leq e \leq 1$. A COR of 0 informs the technical analyst on a weak likelihood of profitable pullback or simply a ranging market. On the other hand, a COR of 1 informs a perfect propensity to a strong pullback expected to at least reach the nearest resistance level. Therefore, in the context of the COR strategy, the closer e is to 1, there would be a higher inclination to perform a trade. Otherwise without

a substantial movement in the underlying asset(s) price, it would not be a profitable trade due to costs such as spread and trading commissions. Note that unlike the original COR formula in Physics, the relative speed after 'collision' is now the denominator instead of the numerator as we are implementing the dip-trip strategy which targets dip buying, typically when RSI drops significantly due to the asset being oversold. As a result, when the RSI drops, e should increase. Conversely, when the asset becomes increasingly overbought, $e \to 0$.

Here, the formula can be interpreted as that the present relative RSI, the closer it is to the previous value the more 'elastic' price action is expected to be to the upside in upcoming time steps with the highest likelihood to reach the nearest resistance level. Then, if $e_t \geq E$ where $0 \leq E \leq 1$ is a tuned threshold parameter, the condition for COR indicator is met in the dip trip strategy. Thus, together with the other conditions with Bollinger Bands and closing prices above moving averages, the dip trip strategy would be executed.

2.4. Competitor Analysis

While we have not found any paper that exactly describes modeling trading signals after the coefficient of restitution for the dip trip strategy, there are related works which modelled after collision dynamics and elastic systems, and from volatility measures for trading signals.

The first competitor stems from Volatility-Managed Portfolios [6], where the authors improve Sharpe ratios and alphas across factors by reducing exposure when volatility is high. However, the performance can attenuate out-of-sample or differ across markets, thus requiring careful validation [7]. In our project, we aim to develop a principled template for volatility-aware sizing overlay on dip-trip entries.

Next, Bergfast introduced a dip-searching swing strategy [8] which directly targets dips with band and range-based logic aligned to our project's entry design framework. However, it is based on a simpler rule set without modern walk-forward and sensitivity quantification. Our proposed framework thus serves as a baseline to benchmark incremental value of volatility-sensitive sizing and exits.

Also from volatility, we have Volatility-Scaled Momentum [12], where volatility targeting reduces crash risk and improves risk-adjusted returns in trend strategies. However, factor-level results may not translate one-for-one to single-name dip entries. In contrast, our project offers a playbook for scaling exposure and risk during high-vol states in the dip-trip strategy.

On the side of physics-inspired approaches, there was work on modeling price action with velocity and mass concepts stemming from the definition of momentum in physics [10]. The paper established relationships of prices as mass particles but in a financial context, and related it to physical momentum but in terms of price action. However, the relationships were not adopted to trading strategies such as the dip trip and remain to be evaluated and enhanced.

In summary, our project is aimed at exploiting buying dip opportunities through "volatility-sensitive sizing and exits" approach adopted from natural occurring events like dropping and bouncing ball, combined with a "walk-forward backtesting" which sets its uniqueness among competing research works, specially by introducing the missing *energy retention* gauge for pullbacks.

2.5. Backtesting Guardrails

The plan to use walk-forward optimization with fixed parameters during testing directly addresses data-snooping concerns raised in classical critiques of technical trading studies, where after rule selection the results often overstate efficacy. Incorporating statistical guardrails inspired by the data-snooping literature and maintaining minimum win rate and per-trade return constraints further reduces the risk of spurious rule discovery and supports credible slope estimation for volatility and profitability sensitivity analysis.

2.6. Transaction Costs Considerations

Trading cost realism materially affects short-horizon strategies, and large-scale institutional evidence indicates that, when measured accurately with live trading data, capacity and net returns of anomalies such as momentum can be substantially larger than once thought, while short-term reversal styles are more constrained by costs [9]. This underscores the importance of modeling slippage and impact explicitly in the backtest and of favoring holding periods and entry and exit mechanics that preserve edge net of costs for swing-horizon dip-trip trades.

2.7. The Data Considerations

Furthermore, our decision to test across US equities and potentially add diversified asset classes is appropriate for assessing sensitivity stability, with the caveat that volatility management effects can vary across markets, suggesting regime identification or stratification by volatility state to improve external validity of the estimated slope [7]. Explicit regime tagging during walk-forward validation (e.g., high-volatility versus low-volatility windows) will thus help determine when volatility increases are beneficial or detrimental for dip-trip profitability.

3. Methodology

Under the scope of this project we seek to solve the problem of *untimely entry* in buying dip strategy. For this, we build a trading algorithm that implements the COR strategy on the selected securities. In this section we provide a full description of the tools and procedures we used to meet such a goal. We present the data selection, the data preprocessing, the implementation procedures of both traditional and the COR methods, and the backtesting procedure.

3.1. Data selection

The selection of securities for swing trading requires rigorous scrutiny in an attempt to avoid the negative implication of slippage over the performance of the strategy. Liquidity is the key factor for this selection as its increase leads to more volatility and consequently more profitable opportunities for swing traders [11].

Under this fact and at this stage of the project, for the sake of algorithm optimization, we retrieved from Yahoo Finance Magnificent Seven (US) stock price data in three timeframes. These tickers are: Apple, Inc. (AAPL), Amazon.com, Inc. (AMZN), Alphabet, Inc. (GOOG), Meta Platforms, Inc. (META), Netflix, Inc. (NFLX), and Tesla, Inc (TSLA). Firstly, we pulled 1 year period bulk of 1 hour, 4 hour and daily price data spanning October 10th 2024 to October 10th 2025 inclusive. Secondly, we retrieved more granular price data, with 30-minute intervals. Due to Yahoo Finance API's restrictions, these shorter frequency datasets could only range September 10th 2025 to October 10th 2025 inclusive for 30 minute data. In addition to the Magnificent Seven stock price data, we collected S&P500 (^GSPC) to be used to compute the COR indicator.

Additionally, for further validation of our hypothesis that extreme volatility could lead to potentially more profitable swing trades, we experimented leveraged ETFs with the same frequencies and respective time horizons as the Yahoo Finance Magnificent Seven stocks. The tickers of choice are: ProShares UltraPro QQQ (Ticker: TQQQ), Direxion Daily Semiconductor Bull 3x Shares (Ticker: SOXL), Direxion Daily TSLA Bull 2X Shares (Ticker: TSLL), Direxion Daily S&P 500 Bull 3x Shares (Ticker: SPXL), GraniteShares 2x Long NVDA Daily ETF (Ticker: NVDL), and Direxion Daily Technology Bull 3x Shares (Ticker: TECL).

3.1.1 Data preprocessing

We performed data cleaning in order to make sure no possible anomalies, such as missing data and outliers, that can negatively impact the performance of the algorithm. Additionally, the data preprocessing step of our dip trip trading algorithm includes the computation of all involved indicators.

3.2. The Traditional Buying Dip Approach

The traditional buying dip approach is employed as a benchmark to evaluate the performance of the COR buying dip strategy that we uniquely develop in this project. Such an approach is deemed to be traditional as it is one of the commonly used by the buying dip practitioners and it consists of a simultaneous use of Relative Strength Index (RSI), Bollinger Bands, and Moving Averages to time their entries in bullish pullbacks of the asset prices.

From the perspective of the benchmark or traditional strategy described above, the buying dip signal will be triggered by a combination of the following circumstances of the asset price that is moving upward after bouncing on the support level:

- The current asset price is above the long-term Simple Moving Average;
- The current asset price is below the 30-level of the RSI indicator, meaning the asset is still undervalued; and
- The current asset price is crossing the lower band of the Bollinger Bands indicator, confirming the asset undervaluation.

3.3. The COR Strategy

The COR buying dip strategy, as developed in this work, combines the traditional buying dip approach and the COR indicator. While the traditional strategy will simply tell the trader what the optimal entry spot or price is, the COR indicator provides information on the strength of the bullish pullback.

Inspired by the Physics Coefficient of Restitution where it indicates the amount of kinetic energy remaining in the bouncing ball right after the collision, as introduced in a prior section above, when the RSI of the asset dips significantly and depending on market conditions via the market's RSI, e tells us how much propensity there is for the asset price to rebound. For example, if the market's RSI before and after remain about the same while a large dip was observed for the asset's RSI, then the asset's price is expected to rebound back to a level aligned to the market's RSI.

As for the market component of the RSI in the formula for COR, we use the S&P 500 index (Ticker: ^GSPC) as an indicator of the broader market. Both asset and market data would have the same frequency and timeframe.

In the implementation of the COR strategy the buy signal is cumulatively triggered by all conditions described in the traditional strategy plus a COR value not below a preset threshold. The COR lower threshold is set in order to make sure that at the entry price the asset has enough likelihood to continue in the profitable bullish movement.

As part of the risk management procedure, the algorithm uses the volatility-adjusted position sizing. Volatility adjustment of the position size involves the use of Average True Range as discussed in the Theoretical Framework. The risk (proportion of portfolio cash) was also included as a hyperparameter for optimization, ranging between 10% and 90%.

3.4. Exit signal generation

Both strategies we consider in this project rely on the same exit approach. The algorithm will trigger the exit signal as one of the following conditions is met:

- The current asset price hits the predefined *take profit* level;
- The current asset price hits the predefined *stop loss* level; or
- The asset movement is experiencing *mean reversion*.

For the purpose of the goals and objectives of this research the take profit level will be set at a predefined threshold not below the halfway to the nearest resistance level from the support zone. On the other hand, the stop loss is configured as a *trailing stop loss* which looks to exit the position close to the reversion price.

3.5. Local backtesting

The plan to use a single-step, out-of-sample chronological split to preserve data integrity and prevent lookahead bias with fixed parameters during testing directly addresses data-snooping concerns raised in classical critiques of technical trading studies, where after rule selection the results often overstate efficacy. This strict chronological train-test split was thus implemented rather than traditional randomized cross-validation. In particular, for each asset, the 1-year historical dataset spanning October 10, 2024, to October 10, 2025, was segmented into two distinct, non-overlapping windows using an optimization split factor of 0.70. The training data spanned approximately October 2024 to late June 2025. This data was utilized to execute an exhaustive grid search across all parameter combinations defined in the optimization grid. The parameter set that maximizes the designated objective function “Total Return” was selected for use in the test dataset, which spanned approximately late June 2025 to October 10, 2025. Note that the optimal parameter set discovered in the training window was locked and evaluated exactly once on this unseen test data to compile the final performance metrics (Total Return, Win Rate, Sharpe Ratio, and Maximum Drawdown) reported in Section 4.

For parameters that form the trading signal and the underlying derivations that lead to specific conditions of the signal, they are added to the optimization grid, each with a range of values for hyperparameter search. As there is a large number of combinations, optimization is performed in parallel across all available CPUs locally.

3.6. Evaluation

Sharpe ratio, total return, and maximum drawdown are performance metrics we will use to evaluate the COR strategy with the traditional approach serving as the comparative benchmark. However, more than just using these metrics to assess the performance of the COR strategy against the traditional approach, they will help evaluate the method across different timeframes.

4. Results

The plots of the Magnificent Seven US stock prices over time, in a 1-day frequency, as shown in Figure 1, denoted the predominance of a smooth bullish tendency along the observation time. Such a behavior appears to be explicitly more expressive for tickers META, MSFT, and TSLA. The observed ETFs, in the same period and frequency, performed in a similar fashion as demonstrated in Figure 2. Additionally, the overall movement for the lower frequencies for both stocks and ETFs followed the tendency of the 1-day frequency price movement.

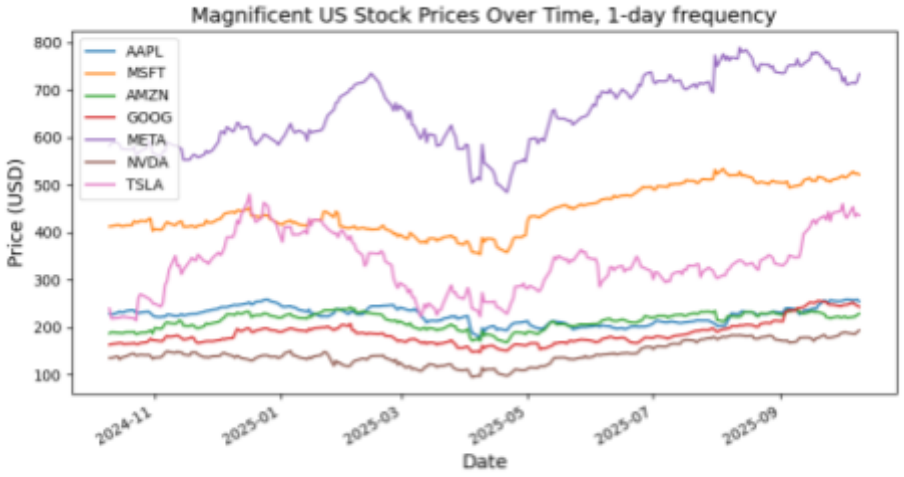


Figure 1: Magnificent Seven US stock prices over the observation time

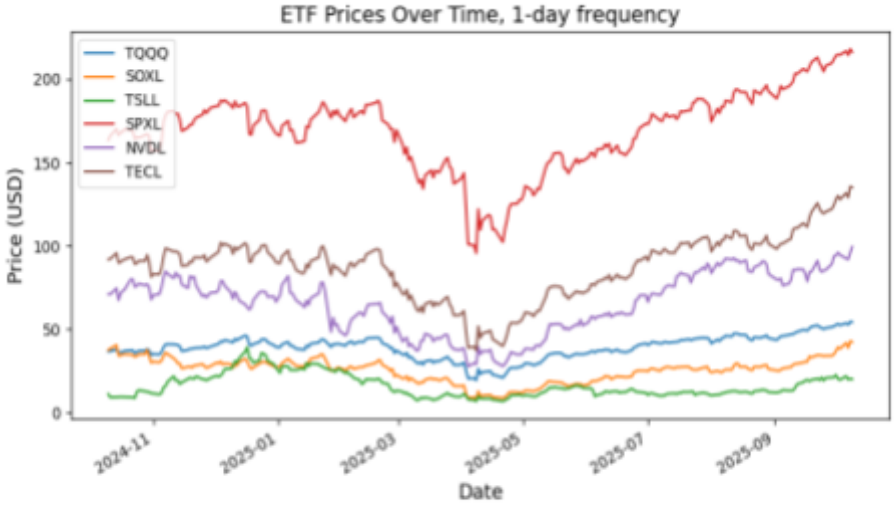


Figure 2: Leveraged ETFs prices over the observation time

Plotting the realized volatility of the studied securities, as shown in Figure 3, we clearly observe that the leveraged ETFs were more volatile than the Magnificent Seven US stocks, an overall behavior for 1-hour, 4-hour, and 1-day data frequencies. TSLA and its tracking ETF, TSLL, were the most volatile securities among stocks and leveraged ETFs, respectively. In addition, the 1-day data frequency was the most volatile among all observed frequencies.

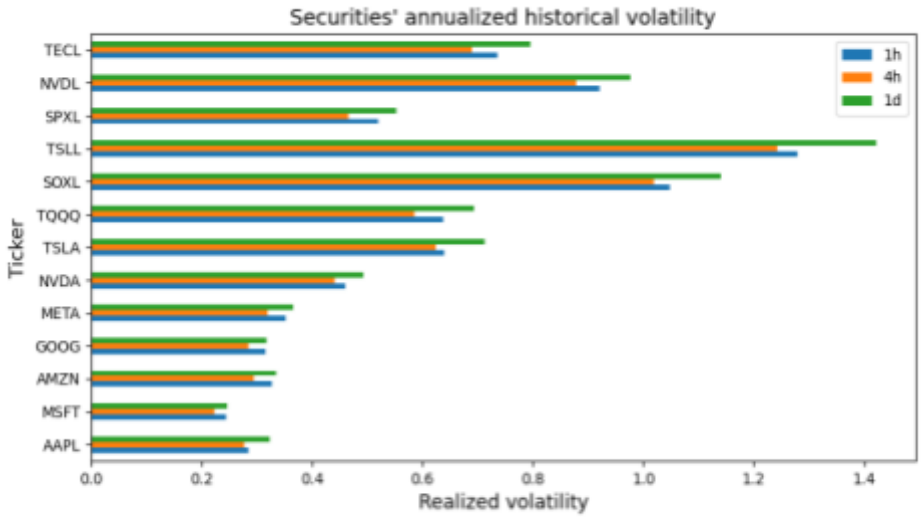


Figure 3: Annualized realized volatility of the observed securities

In our experimentations with backtesting and optimization of the parameters that form the trading signal, a starting cash position of 1,000,000 USD was used for each parameter combination. As the number of parameter combinations is large (in the thousands), for the purpose of formulating the pipelines to perform the optimization so that adequate comparison between methods can be obtained at this stage of the project. Fractional share trading was also enabled. Experiments involving an optimization objective of maximizing total returns and data frequencies of 1 hour, 4 hour and daily were conducted with a timeframe of 1 year, spanning 10th October 2024 to 10th October 2025 inclusive. An exception for the 30-minute interval experiments were made to restrict the past 1 month spanning 10th September 2025 to 10th October 2025 was made due to API limitations.

The assumptions that there are neither trading commissions nor risk-free rates were made. We also made comparisons between the metrics used in the trading signal, to evaluate the tradeoffs between using one or more of the metrics, with the COR metric as the baseline. The abbreviations are as follows:

- BB: Bollinger Bands
- COR: Coefficient of Restitution (Adapted)
- RSI: Relative-Strength Index

Table 1: Results obtained with optimization and fractional trading

METRICS	STOCKS									LEVERAGED ETFs								
	TICKER	With COR (BB, RSI, COR)				Without COR (BB, RSI)				TICKER	With COR (BB, RSI, COR)				Without COR (BB, RSI)			
		30Min	1Hour	4Hours	Daily	30Min	1Hour	4Hours	Daily		30Min	1Hour	4Hours	Daily	30Min	1Hour	4Hours	Daily
Total return (%)	AAPL	-0.43	**1.77**	0	0	-0.4	2.16	0	0	TQQQ	0	**3.56**	12.17	0	0.66	0	12.04	0
Win rate (%)		50	**42.86**	N.A.	N.A.	50	50	N.A.	N.A.		N.A.	**50**	75	N.A.	100	N.A.	75	N.A.
Sarpe ratio		-2.4	**0.38**	N.A.	N.A.	-2.86	0.47	N.A.	N.A.		N.A.	**0.34**	1.81	N.A.	2.73	N.A.	1.8	N.A.
Max. Drawdown		-0.67	**-3.53**	0	0	-0.58	-2.7	0	0		0	**-5.02**	-4.69	0	-0.29	0	-4.79	0
Total trades		2	**7**	0	0	2	4	0	0		0	**8**	4	0	1	0	4	0
Total return (%)	AMZN	0	**-1.61**	0	0	0	-1.41	0	0	SPXL	0	**4.91**	10.97	0	-0.78	6.23	10.97	0
Win rate (%)		N.A.	**54.55**	N.A.	N.A.	N.A.	54.55	N.A.	N.A.		N.A.	**57.14**	80	N.A.	0	66.67	80	N.A.
Sarpe ratio		N.A.	**-0.25**	N.A.	N.A.	N.A.	-0.23	N.A.	N.A.		N.A.	**0.63**	2	N.A.	-3.46	0.8	2	N.A.
Max. Drawdown		0	**-4.28**	0	0	0	-5.13	0	0		0	**-1.75**	-2.64	0	-0.81	-1.75	-2.64	0
Total trades		0	**11**	0	0	0	11	0	0		0	**7**	5	0	1	6	5	0
Total return (%)	GOOG	0.44	**2.9**	7.7	0	-0.52	5.12	0	0	TECL	0	**9.38**	6.39	5.57	1.09	8.13	11.18	0
Win rate (%)		100	**62.5**	100	N.A.	50	66.67	N.A.	N.A.		N.A.	**75**	66.67	100	100	75	75	N.A.
Sarpe ratio		1.73	**0.5**	2.06	N.A.	-1.72	0.75	N.A.	N.A.		N.A.	**0.83**	0.97	2.49	5.06	0.72	1.58	N.A.
Max. Drawdown		-0.21	**-1.94**	-2.54	0	-0.96	-2.11	0	0		0	**-2.41**	-5.34	-0.65	0	-2.41	-5.34	0
Total trades		1	**8**	4	0	2	9	0	0		0	**4**	3	1	1	4	4	0
Total return (%)	META	0	**1.25**	0	2.03	0	1.44	1.56	2.03	SOXL	0	**9.75**	8.78	0	0	9.62	0	0

Win rate (%)		N.A.	**57.14**	N.A.	100	N.A.	66.67	100	100		N.A.	**75**	100	N.A.	N.A.	75	N.A.	N.A.
Sarpe ratio		N.A.	**0.17**	N.A.	1.17	N.A.	0.35	1	1.17		N.A.	**0.68**	2.21	N.A.	N.A.	0.65	N.A.	N.A.
Max. Drawdown		0	**-5.23**	0	-2.04	0	-2.07	-1.22	-2.04		0	**-6.25**	-2.47	0	0	-7.23	0	0
Total trades		0	**14**	0	1	0	3	1	1		0	**4**	1	0	0	4	0	0
Total return (%)		0	**0.33**	-0.46	4.75	0	1.86	4.05	4.75		0	**15.94**	9.06	6.46	0	23.13	-5.9	6.46
Win rate (%)		N.A.	**100**	50	100	N.A.	100	75	100		N.A.	**100**	100	100	N.A.	100	0	100
Sarpe ratio	NFLX	N.A.	**0.19**	-0.15	3.03	N.A.	0.65	1.1	3.03	NVDL	N.A.	**1.31**	1.14	1.38	N.A.	1.74	-1.55	1.38
Max. Drawdown		0	**-1.07**	-3.19	0	0	-1.07	-3.93	0		0	**-2.9**	-5.34	-4.78	0	-1.37	-5.9	-4.78
Total trades		0	**1**	2	1	0	2	4	1		0	**4**	1	1	0	4	1	1
Total return (%)		0.82	**13.59**	6.81	0	0	15.61	12.15	0		0	**15.94**	9.06	6.46	0	23.13	-5.9	6.46
Win rate (%)		100	**83.33**	100	N.A.	N.A.	77.78	100	N.A.		N.A.	**100**	100	100	N.A.	100	0	100
Sarpe ratio	TSLA	1.62	**1.32**	1.71	N.A.	N.A.	1.2	2.09	N.A.	TSLL	N.A.	**1.31**	1.14	1.38	N.A.	1.74	-1.55	1.38
Max. Drawdown		-0.49	**-4.21**	-0.48	0	0	-4.19	-1.27	0		0	**-2.9**	-5.34	-4.78	0	-1.37	-5.9	-4.78
Total trades		1	**6**	1	0	0	9	2	0		0	**4**	1	1	0	4	1	1

From the results in Table 1, there was no clear relationship between the frequency of the data and the results' metrics. Several tickers also had varying frequencies that had zero trades, which contributed to the majority of the 0% maximum drawdowns. The highest total returns were recorded with Tesla (Ticker: TSLA) for the strategy without COR (i.e. only BB and RSI) on the 1 hour frequency, which recorded a 15.61% total returns and a Sharpe ratio of 1.20. However, the strategy with COR yielded high returns of 13.59% with Tesla again, but with a higher win rate of 83.33% and a higher Sharpe ratio of 1.32 compared to the strategy without COR.

Next, we explored the use of ATR to size our positions instead of maximizing the position by allowing fractional trading. This led to the results in Table 2.

Table 2: Results obtained with optimization and ATR to size positions

METRICS	STOCKS									LEVERAGED ETFs								
	TICKER	With COR (BB, RSI, COR)				Without COR (BB, RSI)				TICKER	With COR (BB, RSI, COR)				Without COR (BB, RSI)			
		30Min	1Hour	4Hours	Daily	30Min	1Hour	4Hours	Daily		30Min	1Hour	4Hours	Daily	30Min	1Hour	4Hours	Daily
Total return (%)	AAPL	-0.43	**1.77**	0	0	-0.4	2.16	0	0	TQQQ	0	**3.92**	13.54	0	0.73	0	13.4	0
Win rate (%)		50	**42.86**	N.A.	N.A.	50	50	N.A.	N.A.		N.A.	**50**	75	N.A.	100	N.A.	75	N.A.
Sarpe ratio		-2.4	**0.38**	N.A.	N.A.	-2.86	0.47	N.A.	N.A.		N.A.	**0.34**	1.81	N.A.	2.73	N.A.	1.8	N.A.
Max. Drawdown		-0.67	**-3.53**	0	0	-0.58	-2.7	0	0		0	**-5.57**	-5.2	0	-0.32	0	-5.32	0
Total trades		2	**7**	0	0	2	4	0	0		0	**8**	4	0	1	0	4	0
Total return (%)	AMZN	0	**-1.61**	0	0	0	-1.41	0	0	SPXL	0	**5.45**	12.23	0	-0.87	6.93	12.23	0
Win rate (%)		N.A.	**54.55**	N.A.	N.A.	N.A.	54.55	N.A.	N.A.		N.A.	**57.14**	80	N.A.	0	66.67	80	N.A.
Sarpe ratio		N.A.	**-0.25**	N.A.	N.A.	N.A.	-0.23	N.A.	N.A.		N.A.	**0.63**	2	N.A.	-3.46	0.8	2	N.A.
Max. Drawdown		0	**-4.28**	0	0	0	-5.13	0	0		0	**-1.94**	-2.93	0	-0.9	-1.94	-2.93	0
Total trades		0	**11**	0	0	0	11	0	0		0	**7**	5	0	1	6	5	0
Total return (%)	GOOG	0.44	**2.9**	7.7	0	-0.52	5.12	0	0	TECL	0	**10.44**	7.07	0	1.21	9.04	12.43	6.19
Win rate (%)		100	**62.5**	100	N.A.	50	66.67	N.A.	N.A.		N.A.	**75**	66.67	N.A.	100	75	75	100
Sarpe ratio		1.73	**0.5**	2.06	N.A.	-1.72	0.75	N.A.	N.A.		N.A.	**0.83**	0.97	N.A.	5.06	0.72	1.58	2.49
Max. Drawdown		-0.21	**-1.94**	-2.54	0	-0.96	-2.11	0	0		0	**-2.68**	-5.93	0	0	-2.68	-5.93	-0.72
Total trades		1	**8**	4	0	2	9	0	0		0	**4**	3	0	1	4	4	1

Total return (%)		0	**1.25**	0	2.03	0	1.44	1.56	2.03		0	**10.83**	9.75	0	0	10.69	0	0
Win rate (%)		N.A.	**57.14**	N.A.	100	N.A.	66.67	100	100		N.A.	**75**	100	N.A.	N.A.	75	N.A.	N.A.
Sarpe ratio	META	N.A.	**0.17**	N.A.	1.17	N.A.	0.35	1	1.17	SOXL	N.A.	**0.68**	2.21	N.A.	N.A.	0.65	N.A.	N.A.
Max. Drawdown		0	**-5.23**	0	-2.04	0	-2.07	-1.22	-2.04		0	**-6.93**	-2.73	0	0	-8.01	0	0
Total trades		0	**14**	0	1	0	3	1	1		0	**4**	1	0	0	4	0	0
Total return (%)		0	**0.33**	-0.46	4.75	0	1.86	4.05	4.75		0	**17.81**	10.07	7.18	0	25.93	-6.56	7.18
Win rate (%)		N.A.	**100**	50	100	N.A.	100	75	100		N.A.	**100**	100	100	N.A.	100	0	100
Sarpe ratio	NFLX	N.A.	**0.19**	-0.15	3.03	N.A.	0.65	1.1	3.03	NVDL	N.A.	**1.31**	1.14	1.39	N.A.	1.74	-1.55	1.39
Max. Drawdown		0	**-1.07**	-3.19	0	0	-1.07	-3.93	0		0	**-3.22**	-5.92	-5.3	0	-1.52	-6.56	-5.3
Total trades		0	**1**	2	1	0	2	4	1		0	**4**	1	1	0	4	1	1
Total return (%)		0.82	**13.59**	6.81	0	0	15.61	12.15	0		1.71	**41.56**	0	0	0	43.77	27.98	0
Win rate (%)		100	**83.33**	100	N.A.	N.A.	77.78	100	N.A.		100	**85.71**	N.A.	N.A.	N.A.	77.78	100	N.A.
Sarpe ratio	TSLA	1.62	**1.32**	1.71	N.A.	N.A.	1.2	2.09	N.A.	TSLL	1.54	**1.51**	N.A.	N.A.	N.A.	1.35	2.11	N.A.
Max. Drawdown		-0.49	**-4.21**	-0.48	0	0	-4.19	-1.27	0		-1.17	**-9.35**	0	0	0	-9.29	-2.76	0
Total trades		1	**6**	1	0	0	9	2	0		1	**7**	0	0	0	9	2	0

From the results in Table 2, it was observed that similarly, there was no clear relationship between the timeframe of the closing prices and the results' metrics. In particular, the timeframe of daily for Netflix (Ticker: NFLX) yielded the best Sharpe ratio of 3.03 for both strategies. However, the best recorded total returns was with the 1 hour frequency data with Tesla (Ticker: TSLA), which also yielded a relatively high Sharpe ratio of 1.20 and highest total returns of 17.44% for the strategy without COR. However, the strategy for COR yielded not only the second highest return but also a higher win rate of 83.33%, and a higher Sharpe ratio of 1.32. At this point, it was observed that having ATR to size positions during trades increases win rate and total returns for the stocks. In particular, having COR was found to have increased the win-rate by a notable amount alongside Sharpe ratio, while sacrificing a small percentage of returns. However, since the best returns were observed for the 1-hour interval instead of higher frequencies, it potentially suggests a local maxima for the frequency-volatility-total returns tradeoff. A notable observation was that returns were seemingly greater for stocks with greater historical volatilities such as Tesla. This effect was found to be even more prominent with ATR.

Next, to further validate our hypothesis that extreme volatility could potentially reap more rewards, leveraged ETFs were also experimented with for the same frequencies and respective time horizons. The tickers of choice are: ProShares UltraPro QQQ (Ticker: TQQQ), Direxion Daily Semiconductor Bull 3x Shares (Ticker: SOXL), Direxion Daily TSLA Bull 2X Shares (Ticker: TSLL), Direxion Daily S&P 500 Bull 3x Shares (Ticker: SPXL), GraniteShares 2x Long NVDA Daily ETF (Ticker: NVDL), and Direxion Daily Technology Bull 3x Shares (Ticker: TECL).

From the results in Table 1 and 2, a similar result on having ATR yielding higher total returns was observed. Moreover, leveraged ETFs have shown to yield significantly higher total returns, and even much more with ATR as shown in Table 2. Our hypothesis that higher volatility, moreover so for the same stock ticker but with leverage, could potentially reap higher returns rewards appeared to be largely validated by the results in Tables 1 and 2. For example, Direxion Daily TSLA Bull 2X Shares (Ticker: TSLL) which has the underlying assets in the ETF to track the performance of Tesla stock, using ATR yielded significantly higher returns of 43.77% without COR and 41.56% with COR in Table 2, as compared to the same 1-hour frequency for Tesla (Ticker: TSLA), which yielded 17.44% without COR and 15.16% total returns with COR in Table 2. However, it was noted that the maximum drawdowns also significantly increased to -9.35% from -4.67%, in the case of using COR with ATR. This result showed that including leverage could exacerbate total returns, but also the drawdowns. Other leveraged ETFs in Table 1 and 2 also showed much higher returns than non-leveraged assets in Tables 1 and 2.

Our project code is hosted on a GitHub repository, which can be accessed via the link at https://github.com/weihao94/WQU-MScFE-690-Capstone-Group-11206.

5. Discussion

From our initial optimization results with a stock ticker from the magnificent seven companies in the US, it was found that there was no clear relationship between higher intervals of asset prices and a higher total return and win rate. Moreover, with ATR for position sizing in trades, the total returns, win rates and Sharpe ratios were found to be higher for the same asset and frequency. These results thus indicate the importance of ATR in risk adjustments to our trades. Our results also showed that having COR in the strategy helped increase the win-rate and Sharpe ratio without sacrificing much returns, which indicated that our proposed COR strategy is effective in 'securing' wins when added to the vanilla strategy involving the BB and RSI. Moreover, there are a notably small number of trades during the evaluation period, which places results with high (or perfect) win-rates with caution due to the low number of trades to evaluate the effectiveness of our strategy.

We have also established a relationship with higher volatility assets (e.g. Tesla stock) having greater potential for returns, especially with ATR. This volatility relationship with returns was found to be exacerbated with leverage, which was validated with the leveraged ETF for Tesla (Ticker: TSLL), with and without ATR. As currently we have only evaluated a stock's performance with a leveraged version of its asset, to further extend our hypothesis and results, we would thus expand the comparisons in future to more pairs of assets and their leveraged versions. This would help to further investigate potential relationships between volatility of assets with their total returns, with and without ATR. Combinations of metrics will likewise also be sequentially evaluated starting with our proposed COR metric, adding in more like Bollinger Bands, RSI, leading to a combination of all metrics to determine the trading signal.

6. Conclusion

Our research successfully demonstrated the potential of physics-inspired technical analysis through the development and implementation of the Coefficient of Restitution (COR) indicator for swing trading applications. The proposed COR strategy addresses a fundamental limitation in traditional dip-buying approaches by providing quantitative assessment of pullback strength, thereby improving entry timing and trade quality evaluation. The primary contribution of this work resides in the novel adaptation of collision dynamics principles to financial market analysis. By translating the physics concept of energy retention after collision (i.e. COR) into a metric for measuring price momentum recovery, the COR indicator fills a critical gap in existing technical analysis frameworks. The mathematical formulation in (12) successfully captures the relationship between asset-specific price movements and broader market conditions, providing traders with actionable insights about potential trade outcomes.

Furthermore, our empirical analysis reveals several important insights. Firstly, the integration of ATR-based position sizing significantly enhances performance metrics across all timeframes, demonstrating the value of volatility-aware risk management. Secondly, more volatile assets with leverage consistently produced superior returns and risk-adjusted performance, with the 1-hour interval achieving exceptional results including 85.71% win rates and positive Sharpe ratios. In particular, with ATR, the 1-hour interval yielded the highest total return of 41.56% with COR and 43.77% without COR, and a high Sharpe ratio of 1.51 with COR and 1.35 without COR. Lastly, the COR strategy's performance improvements over traditional methods validate the theoretical foundation and practical applicability of physics-inspired indicators in financial markets.

Overall, while our initial results appear promising, there are several limitations which warrant acknowledgment. The current analysis focuses primarily on selected stocks from the magnificent seven in the US and selected leveraged ETFs, which are large companies and funds with highly liquid assets during specific market conditions, potentially limiting

generalizability. As a next step, we seek to expand the universe of assets to include diverse sectors, asset classes, historical volatilities and market regimes to validate the COR indicator's robustness across varying conditions. More historical data would also be extracted for each frequency, subject to the limits of Yahoo Finance's API, which in our frequencies contexts are 60 days for 30-minute data, 2 years for hourly data, and years (non-stated limit) for daily data. Moreover, our study period's predominantly bullish market conditions may have favored dip-buying strategies. Additional testing during known bear markets and high-volatility regimes will be crucial for comprehensive strategy validation. The integration of regime identification algorithms could further enhance the COR strategy's adaptability to changing market conditions.

7. Data and Code Availability Statement

To ensure replicability, transparency, and the academic integrity of this study, all computation logic, parameter grids, and backtesting frameworks have been made fully open-source. The complete Python implementation scripts, datasets used in the experiments and their associated results are publicly hosted on our GitHub repository at: https://github.com/weihao94/WQU-MScFE-690-Capstone-Group-11206.

The empirical analysis relied entirely on historical market data fetched from Yahoo Finance. The asset universe comprised of two classes and a market benchmark:

- **Single Equities**: Six selected assets from the US Magnificent Seven companies, including Apple, Inc. (AAPL), Amazon.com, Inc. (AMZN), Alphabet, Inc. (GOOG), Meta Platforms, Inc. (META), Netflix, Inc. (NFLX), and Tesla, Inc. (TSLA).
- **Leveraged ETFs**: ProShares UltraPro QQQ (TQQQ), Direxion Daily Semiconductor Bull 3x Shares (SOXL), Direxion Daily TSLA Bull 2X Shares (TSLL), Direxion Daily S&P 500 Bull 3x Shares (SPXL), GraniteShares 2x Long NVDA Daily ETF (NVDL), and Direxion Daily Technology Bull 3x Shares (TECL).
- **Market Benchmark**: The S&P 500 Index (^GSPC) serves as the baseline proxy for broader market conditions to evaluate the relative RSI differentials within the COR metric.

Due to the native data distribution limitations of the Yahoo Finance API, two distinct time horizons were utilized:

1. **1-Hour, 4-Hour, and Daily Frequencies**: A full 1-year historical window spanning from October 10, 2024, to October 10, 2025, inclusive.
2. **30-Minute Frequency**: A truncated 1-month historical window spanning from September 10, 2025, to October 10, 2025, inclusive, enforced by API structural limits on high-frequency intraday data retrieval.

The backtesting validation framework initiated with a baseline cash position of $1,000,000 USD. To maximize market exposure and optimize capital efficiency during trade execution, fractional share trading was enabled. Lastly, to streamline the hyperparameter search and isolate the mathematical edge of the technical signals, the backtest operated under the assumptions that transaction costs, execution commissions, slippage, and risk-free rates were zero.

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9. Disclaimer

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Appendix

A. Code Repository

The following link is to a GitHub repository containing our project code, datasets used and results: https://github.com/weihao94/WQU-MScFE-690-Capstone-Group-11206.